# Topic 1 — Physics Fundamentals, Measurement and Instruments

**Direct Syllabus Mapping**: General Science — Physical Quantities, Units, Measurement Systems, Dimensions, Standards, Practical & Astronomical Units, Scientific Measuring Instruments and Apparatus.
**Key Must-Score Focus**: SI base units (2019 BIPM definitions via fundamental physical constants), 22 coherent derived units with special names, Astronomical Units (Parsec, Light-Year, AU), Pressure conversions (Bar, Pascal, Torr, psi, kgf/cm²), Specialized units (Dobson, Cusec, NTU, Barrel of oil, Horsepower), and Exhaustive Scientific Instruments Directory (Pyrometer, Pyrheliometer, Anemometer, Tachometer, Fathometer, SONAR, RADAR, Polygraph, Accelerometer, Ringelmann Scale, etc.).

## Current Affairs

- **BIPM 2019 SI Redefinition (Effective May 20, 2019)**: All 7 base units are now anchored to fixed numerical values of fundamental physical constants:
- Kilogram ($kg$) is locked to Planck constant $h = 6.62607015 \times 10^{-34}\ \text{J} \cdot \text{s}$ ($kg \cdot m^2 \cdot s^{-1}$), terminating the physical Platinum-Iridium prototype cylinder (*Le Grand K*) kept at BIPM Sèvres (near Paris).
- Second ($s$) is anchored to caesium frequency $\Delta\nu_{\text{Cs}} = 9,192,631,770\ \text{Hz}$ (unperturbed ground-state hyperfine transition of Cs-133).
- Metre ($m$) is anchored to speed of light in vacuum $c = 299,792,458\ \text{m/s}$.
- Ampere ($A$) is locked to elementary charge $e = 1.602176634 \times 10^{-19}\ \text{C}$ ($1\ \text{A} = 6.2415 \times 10^{18}\ \text{electrons/second}$).
- Kelvin ($K$) is locked to Boltzmann constant $k = 1.380649 \times 10^{-23}\ \text{J/K}$ ($kg \cdot m^2 \cdot s^{-2} \cdot K^{-1}$), eliminating reliance on the triple point of water ($273.16\ \text{K}$).
- Mole ($mol$) contains exactly $6.02214076 \times 10^{23}$ specified elementary entities (Avogadro constant $N_A$).
- Candela ($cd$) is locked to luminous efficacy $K_{\text{cd}} = 683\ \text{lm/W}$ for monochromatic radiation of frequency $540 \times 10^{12}\ \text{Hz}$.
- **New SI Prefixes (27th CGPM, Versailles, 2022)**:
- Large: **Ronna** ($R = 10^{27}$), **Quetta** ($Q = 10^{30}$). Mass of Earth $\approx 6\ \text{ronnagrams}$ ($6 \times 10^{24}\ \text{kg} = 6 \times 10^{27}\ \text{g}$).
- Small: **Ronto** ($r = 10^{-27}$), **Quecto** ($q = 10^{-30}$). Mass of electron $\approx 1\ \text{rontogram}$.
- **Solar Constant & TSIS-1 (NASA Consensus)**: Total Solar Irradiance (TSI) received at $1\ \text{AU}$ outside Earth's atmosphere is standardized at $1361.6 \pm 0.3\ \text{W/m}^2 \approx 1.4\ \text{kW/m}^2$ (measured by TSIS-1 / SORCE; varies by $\sim 0.1\%$ over the 11-year solar sunspot cycle).
- **MEMS Accelerometers in Modern Tech (IAS 2023 Fact)**: Accelerometers sense dynamic acceleration and tilt. Crucial for: (1) Airbag deployment via instant deceleration detection, (2) Free-

fall sensor (FFS) in laptops parking HDD heads to prevent drive crash, (3) Display auto-rotation between portrait and landscape in smartphones.

## 1.1 Physical Quantities, Systems of Units & Dimensions

- **Physical Quantity**: Any property of a material or system that can be quantified and measured with an instrument.

$$\text{Physical Quantity} = \text{Numerical Value } (n) \times \text{Unit } (u)$$

Since the magnitude is constant: $n_1 u_1 = n_2 u_2$. As unit size increases, numerical value decreases proportionally.

- **Fundamental vs Derived Quantities**:
- *Fundamental (Base)*: Independent quantities that cannot be defined in terms of other physical quantities (7 in SI).
- *Derived*: Quantities formed by multiplying or dividing base units with integer/fractional powers.
- **Historical Unit Systems**: 1. *CGS System* (Gaussian metric): Centimetre (length), Gram (mass), Second (time). 2. *FPS System* (British Imperial): Foot (length), Pound (mass; $1\ \text{lb} = 453.59237\ \text{g}$), Second (time). 3. *MKS System* (Giorgi metric): Metre (length), Kilogram (mass), Second (time). 4. *SI System* (1960, 11th CGPM): Comprehensive rationalized MKS system incorporating 7 base and 2 supplementary units.

## 1.2 Measurement of Length, Mass, Time & Astronomical Distances

- **Julian Year in Astronomy**: Time measurement unit defined as exactly $365.25$ **days of** $86,400$ **SI seconds each** ($= 31,557,600$ s). A light-year is the vacuum travel distance of light in one Julian year:

$$\text{Distance} = c \times t = 299,792,458\ \text{m/s} \times 31,557,600\ \text{s} = 9.460730472 \times 10^{15}\ \text{m} \approx 9.461 \times 10^{15}\ \text{m}$$

- **Parsec Geometry**:

$$\text{Angle } \theta = \frac{\text{Arc}}{\text{Radius}} \implies 1\ \text{arcsecond} = \frac{1\ \text{AU}}{1\ \text{Parsec}}$$

Since $1\ \text{arcsecond} = \frac{1}{3600}^\circ = \frac{\pi}{180 \times 3600}\ \text{radians} = \frac{\pi}{648{,}000}\ \text{rad}$:

$$1\ \text{Parsec} = \frac{1\ \text{AU}}{\theta} = \frac{648,000}{\pi}\ \text{AU} \approx 206,264.8\ \text{AU} \approx 3.0857 \times 10^{16}\ \text{m} \approx 3.262\ \text{light-years}$$

## 1.3 Temperature Scales, Thermometry & Pyrometry

- **Liquid Thermometers**:
- *Mercury Thermometer*: Usable range from $-39^\circ C$ (mercury freezing point) to $357^\circ C$ (boiling point). Can be extended to $500^\circ C$ by filling nitrogen gas under pressure above mercury column.

- *Alcohol Thermometer*: Usable down to $-112^\circ C$ (ideal for arctic/cryogenic regions where mercury freezes).
- **Total Radiation Pyrometer (High Temperature $> 1500^\circ C$)**:
- Based on **Stefan-Boltzmann Law**: Total radiant energy ($E$) emitted per unit surface area of a blackbody per second is directly proportional to the fourth power of its absolute thermodynamic temperature:

$$E = \sigma T^4 \quad (\sigma = 5.670374 \times 10^{-8}\ \mathrm{W \cdot m^{-2} \cdot K^{-4}})$$

- Since it detects thermal radiation at a distance, it never contacts the source. Used for metallurgical blast furnaces, molten steel, kiln chambers, and stellar bodies.

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## 1.4 Measuring Instruments & Modern Experimental Tools

- **SONAR (Sound Navigation and Ranging)**:
- Emits high-frequency **ultrasonic acoustic waves** into water.
- Velocity of sound in seawater is $\approx 1500\ \mathrm{m/s}$ (much faster than $343\ \mathrm{m/s}$ in air).
- Distance $d = \frac{v \times t}{2}$ where $t$ is round-trip echo return time.
- *Active Sonar*: Transmits acoustic ping and detects reflected echo.
- *Passive Sonar*: Does not emit sound; only listens to acoustic signatures of other vessels (submarine stealth).
- **RADAR (Radio Detection and Ranging)**:
- Uses pulsed **radio waves / microwaves** ($c = 3 \times 10^8\ \mathrm{m/s}$).
- Detects aircraft, missiles, ships, and weather precipitation. Doppler radar additionally measures target radial velocity.
- **Polygraph (Lie Detector)**:
- Invented by John Augustus Larson (1921). Simultaneously tracks 4 autonomic physiological parameters:
  1. Blood pressure (pneumatic cuff), 2. Pulse rate (plethysmograph), 3. Respiration rate (pneumograph tubes), 4. Galvanic Skin Resistance / GSR (sweat gland activation via electrodermal sensors).
- **Ringelmann Smoke Chart**:
- Developed by Maximilien Ringelmann (1888). 5 grids of black lines on white paper representing smoke density from Card 0 ($0\%$ clear) to Card 5 ($100\%$ pitch black). Standard regulatory scale for chimney emission opacity.

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## 1.5 Kinematics Essentials, Average Speed & Impulse Mechanics

- **Average Speed vs Average Velocity**:
- **Equal Distance Segments (Harmonic Mean)**: When a body covers the first half of distance $d$ at speed $v_1$ and second half at speed $v_2$, total distance is $2d$ and total time is $t = \frac{d}{v_1} + \frac{d}{v_2} = d\left(\frac{v_1+v_2}{v_1 v_2}\right)$

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$$v_{\text{avg}} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{2d}{d\left(\frac{v_1+v_2}{v_1 v_2}\right)} = \frac{2v_1 v_2}{v_1+v_2}$$

*(Note: For three equal distances at $v_1, v_2, v_3$: $v_{\text{avg}} = \frac{3v_1v_2v_3}{v_1v_2+v_2v_3+v_3v_1}$).*

- **Equal Time Intervals (Arithmetic Mean)**: When a body travels for time $t$ at speed $v_1$ and next equal time $t$ at speed $v_2$:

$$v_{\text{avg}} = \frac{v_1 t + v_2 t}{2t} = \frac{v_1+v_2}{2}$$

- **Impulse ($J$ or $I$) & Linear Momentum ($p$)**:
- **Definition**: Impulse is the total change in linear momentum produced by a large force acting over a short time interval:

$$\vec{J} = \vec{F} \cdot \Delta t = \Delta \vec{p} = m\vec{v}_f - m\vec{v}_i$$

- **Units**: Newton-second ($N \cdot s$) or kilogram metre per second ($kg \cdot m/s$).
- **Dimensional Formula**:

$$[J] = [F] \cdot [T] = [M^1 L^1 T^{-2}] \cdot [T^1] = [M^1 L^1 T^{-1}]$$

*(Crucial Exam Fact: Impulse and Linear Momentum have the EXACT same dimensional formula $[MLT^{-1}]$ and identical base units).*

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## 1.6 Heat vs Temperature Distinctions, Optics & CGS Conversion Factors

- **Heat (Energy) vs Temperature (Thermal State)**:
- **Heat** is a form of energy transferred across a boundary due to temperature gradient.
  - *Units of Heat*: Joule ($J$ — SI unit), Calorie ($1\ \text{cal} = 4.184\ \text{J}$), Kilocalorie ($1\ \text{kcal} = 1000\ \text{cal} = 4184\ \text{J}$), Erg ($1\ \text{J} = 10^7\ \text{ergs}$), British Thermal Unit ($1\ \text{BTU} \approx 1055\ \text{J} \approx 252\ \text{cal}$).
- **Temperature** is the measure of average kinetic energy of constituent particles.
  - *Units of Temperature*: Kelvin ($K$ — SI base unit), Degree Celsius / Centigrade ($^\circ C$), Degree Fahrenheit ($^\circ F$).
  - *Trap Alert*: **Centigrade / Celsius is a unit of temperature, NOT heat!**
- **Power of an Optical Lens**:
- Defined as reciprocal of focal length ($f$) measured in metres:

$$P = \frac{1}{f\ (\text{in metres})}$$

- **Unit**: Diopter ($D$ or $m^{-1}$).
- Sign convention: Convex (converging) lens has positive focal length $\implies +D$; Concave (diverging) lens has negative focal length $\implies -D$.
- **Small Length Unit Conversions in Centimetres ($cm$)**:

- $1$ micron $(\mu m) = 10^{-6}$ m $= 10^{-4}$ cm $= 10^{-3}$ mm.
- $1$ nanometre $(nm) = 10^{-9}$ m $= 10^{-7}$ cm $= 10$ \AA.
- $1$ Angstrom $(\AA) = 10^{-10}$ m $= 10^{-8}$ cm $= 0.1$ nm.
- $1$ Fermi / Femtometre $(fm) = 10^{-15}$ m $= 10^{-13}$ cm (Smallest unit of length).
- **Pressure and Force Equivalents**:
- $1$ Pascal $(Pa) = 1$ N/m$^2$.
- $1$ bar $= 10^5$ Pa $= 10^6$ dyn/cm$^2 \approx 1.0197$ kgf/cm$^2 \approx 1.0$ bar $\approx 1$ kgf/cm$^2$ ( $1$ kgf/cm$^2 = 98,066.5$ Pa).
- $1$ Newton $= 10^5$ dynes.
- $1$ Joule $= 10^7$ ergs.

## Bilingual Terminology -- Physics Fundamentals and Measurement

| English Term | Hindi Term | Key Anchor |
|---|---|---|
| **Physical Quantity** (भौतिक राशि) | भौतिक राशि | Measurable property; scalar (magnitude only) or vector (magnitude + direction) |
| **SI Unit** (अंतर्राष्ट्रीय मात्रक प्रणाली) | अंतर्राष्ट्रीय मात्रक | International System of Units; 7 base units |
| **Scalar** (अदिश) | अदिश राशि | Has magnitude only; e.g., mass, temperature, speed, energy |
| **Vector** (सदिश) | सदिश राशि | Has magnitude + direction; e.g., velocity, force, acceleration |
| **Length** (लम्बाई) | लम्बाई | SI unit = metre (m) |
| **Mass** (द्रव्यमान) | द्रव्यमान | SI unit = kilogram (kg); measure of matter |
| **Time** (समय) | समय | SI unit = second (s); Cs-133 atomic clock |
| **Electric Current** (विद्युत धारा) | विद्युत धारा | SI unit = ampere (A) |
| **Temperature** (ताप) | ताप | SI unit = kelvin (K); 0 K = absolute zero = -273.15 degC |
| **Luminous Intensity** (ज्योति तीव्रता) | ज्योति तीव्रता | SI unit = candela (cd) |
| **Amount of Substance** (पदार्थ की मात्रा) | पदार्थ की मात्रा | SI unit = mole (mol); Avogadro's number = 6.022 x 10^23 |
| **Parsec** (पारसेक) | पारसेक | Largest astronomical distance unit; 1 parsec = 3.26 light-years |
| **Light-Year** (प्रकाश वर्ष) | प्रकाश वर्ष | Distance light travels in 1 year; unit of distance NOT time |
| **Astronomical Unit (AU)** (खगोलीय इकाई) | खगोलीय इकाई | Mean Earth-Sun distance = 1.496 x 10^11 m |
| **Dobson Unit** (डोबसन इकाई) | डोबसन इकाई | Measures ozone layer thickness; normal = 300 DU; hole < 220 DU |
| **Hertz** (हर्ट्ज) | हर्ट्ज | SI unit of frequency; 1 Hz = 1 cycle/second |

| English Term | Hindi Term | Key Anchor |
|---|---|---|
| **Pascal** (पास्कल) | पास्कल | SI unit of pressure; 1 Pa = 1 N/m2 |
| **Decibel** (डेसिबल) | डेसिबल | Logarithmic unit of sound intensity; 0 dB = threshold of hearing |

## One-Liner Revision -- Physics Fundamentals and Measurement

| # | Fast Fact |
|---|---|
| 1 | SI base units = **7**: m, kg, s, A, K, mol, cd |
| 2 | Supplementary units = Radian (plane angle) + Steradian (solid angle) |
| 3 | New SI prefixes (2022) = Ronna (10^27), Quetta (10^30), Ronto (10^-27), Quecto (10^-30) |
| 4 | Light-Year = unit of **distance** (NOT time); 1 ly = 9.46 x 10^15 m |
| 5 | 1 Parsec = 3.262 light-years = 206,265 AU (coined: Herbert Hall Turner, 1913) |
| 6 | 1 AU = 1.496 x 10^11 m = ~150 million km (average Earth-Sun distance) |
| 7 | Smallest unit of length = **Fermi** (fm = 10^-15 m); used for nuclear radii |
| 8 | Angstrom = 10^-10 m; used for atomic radii and wavelengths |
| 9 | 1 Barrel of oil = **159 litres** (~42 US gallons) |
| 10 | 1 Nautical mile = **1.852 km**; Knot = 1 nautical mile/hour |
| 11 | 1 Cusec = 1 ft3/s = 28.317 litres/s (river/canal flow rate) |
| 12 | Normal atmospheric pressure = **1 atm = 101,325 Pa = 760 mmHg = 760 torr** |
| 13 | 1 bar = 10^5 Pa; 1 millibar = 100 Pa (used in meteorology) |
| 14 | 1 Horsepower = **746 Watts**; Metric HP = 735.5 W |
| 15 | 1 kWh = **3.6 x 10^6 Joules** (BOT unit; on electricity bill) |
| 16 | Dobson Unit: Normal ozone = 300 DU; Ozone hole < **220 DU** |
| 17 | Pyrometer = measures very high temperatures (>1500 degC) without contact |
| 18 | Anemometer = measures **wind speed**; Wind Vane = measures wind **direction** |
| 19 | Barometer: sudden drop = approaching storm; gradual fall = rain; rise = fair weather |
| 20 | Hygrometer = measures **humidity**; Hydrometer = measures **specific gravity of liquids** |
| 21 | Lactometer = checks **milk purity** (density); Butyrometer = measures **fat in milk** |
| 22 | SONAR = uses **ultrasonic waves** (>20 kHz); for submarine/ocean depth detection |
| 23 | RADAR = uses **radio/microwaves**; detects aircraft, ships, weather |
| 24 | Fathometer = measures ocean depth; 1 fathom = 6 feet = 1.83 m |

| # | Fast Fact |
|---|---|
| 25 | ECG = heart; EEG = brain; EMG = muscles; Polygraph = lie detector |
| 26 | 1 kg defined by Planck constant h = 6.626 x 10^-34 J.s (since 2019 BIPM) |
| 27 | Cs-133 atomic clock = defines the **second** (9,192,631,770 transitions) |
| 28 | Kelvin 0 = absolute zero = -273.15 degC (theoretical minimum temperature) |
| 29 | Speed of light (c) = 299,792,458 m/s (defines the metre) |
| 30 | Mole = 6.02214076 x 10^23 entities (Avogadro's number; defines the mole) |